

Key words And Terms

wave function

probability amplitude

probability density

wave of probability

matrix

time-independent state

potential well

波函数

几率幅

几率密度

几率波

矩阵

定态

势阱

Chapter 23 Quantum mechanic

23.1 Wave function and Born's interpretation

23.2 The Heisenberg uncertainty principle

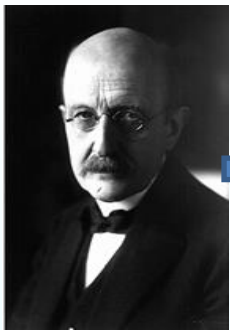
23.3 The Schrödinger equation

23.4 Particles in an infinitely deep square well

23.5 Schrödinger equation for hydrogen atom

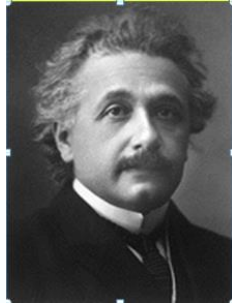
23.6 Periodic table of elements

Planck



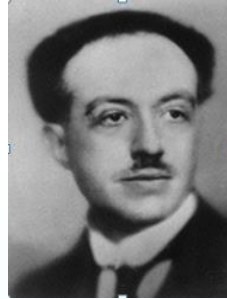
Quantum hypothesis (1900)

Einstein



Photon theory of light (1905)

de Broglie



Matter wave (1923)

Erwin Schrödinger



Wave quantum mechanics (1926)

Bohr

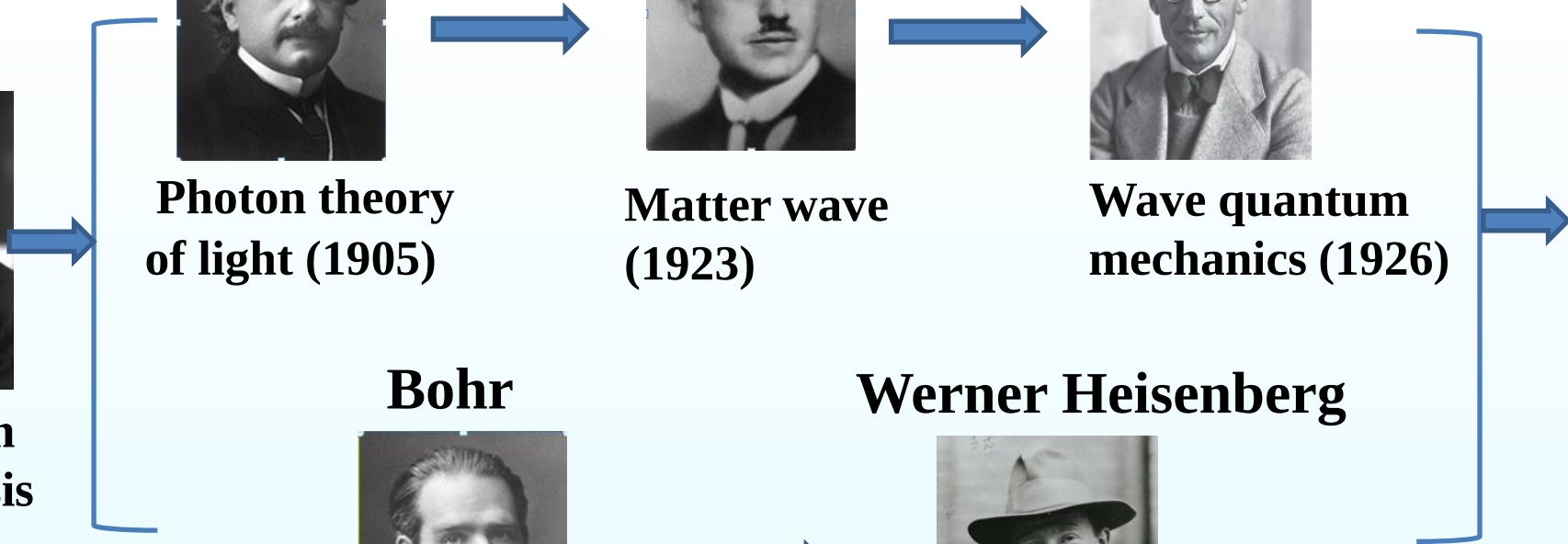


Bohr's model about hydrogen atom (1913)

Werner Heisenberg



Matrix quantum mechanics (1925)



23.1 Wave function and Born's interpretation

微观粒子的运动状态



波函数

描述微观粒子运动基本方程



薛定谔方程

1 .wave function

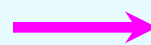
Wave function for a free particle in one dimension

一个沿 x 轴正向传播的频率为 f 的平面简谐波:

$$y = A \cos 2\pi \left(ft - \frac{x}{\lambda} \right)$$

用复数形式表示:

$$y = Ae^{-i2\pi \left(ft - \frac{x}{\lambda} \right)}$$



Take Real
part(实部)

波的强度

$$I \propto A^2$$

对于动量为 P 、能量为 E 的一维自由微观粒子，根据德布罗意假设，其物质波的波函数相当于单色平面波，类比可写成：

$$\Psi(x, t) = Ae^{-i2\pi\left(ft - \frac{x}{\lambda}\right)}$$

$$E = hf \quad p = \frac{h}{\lambda}$$

$$\Psi(x, t) = \Psi_0 e^{-\frac{i}{\hbar}(Et - px)}$$

One dimension

Ψ : represents the displacement as a function of time and position, of a new kind of field --“a matter” field or a matter wave.

Three dimensions

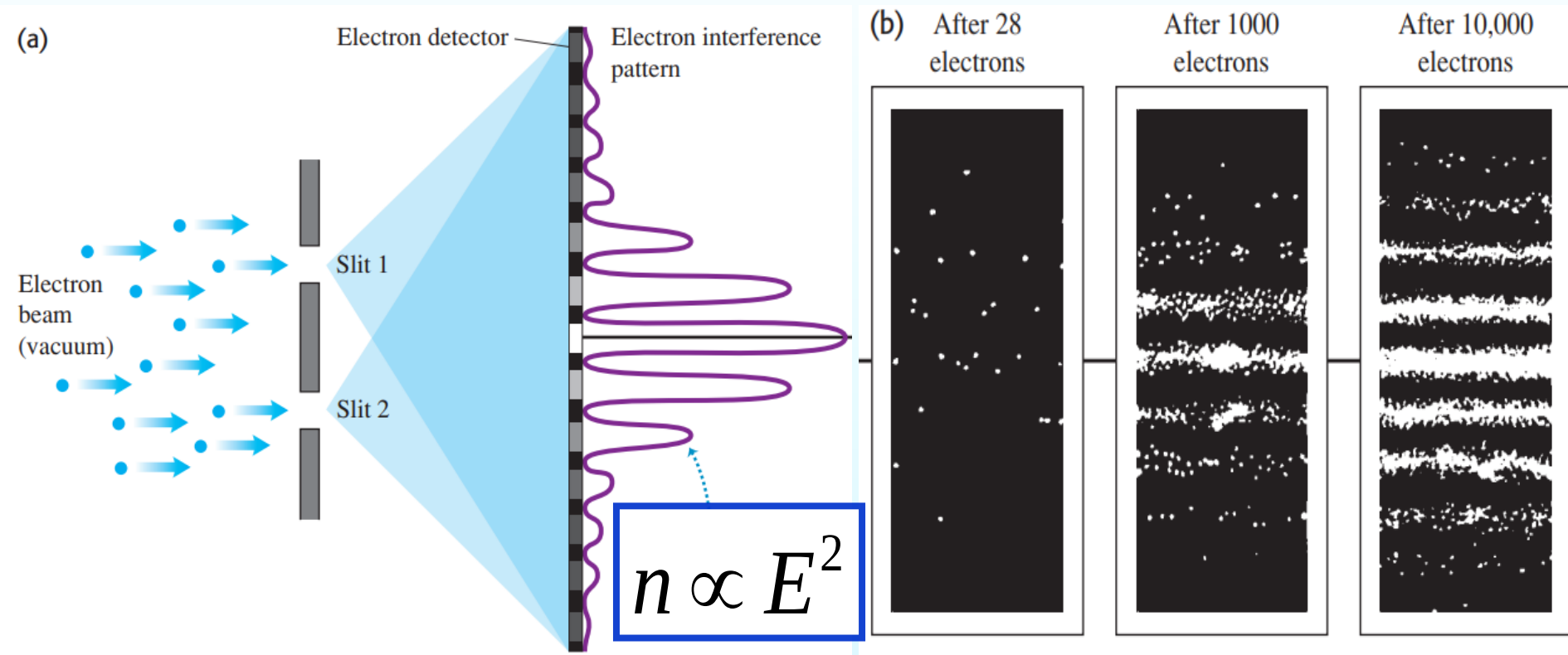
$$\Psi(\vec{r}, t) = \Psi_0 e^{-\frac{i}{\hbar}(Et - \vec{p} \cdot \vec{r})}$$

2. Born's interpretation for wave function

Double-slit interference of electrons

Wave point: The intensity I of any wave is proportional to the square of the displacement amplitude. $I \propto E^2$

Particle point: The energy density of a light beam is proportional to the number of photons per unit volume. $I \propto n$



Born proposed that the wave function amplitude is related to the **probability** of finding the particle within a unit volume at that location, and the exact value of the probability is determined by Ψ^2
— **wave of probability**

Probability density 概率密度



$$|\Psi(r, t)|^2 = \Psi\Psi^* = \Psi_0^2$$



Max Born
(1882–1970),
1954 “Noble”

在某处德布罗意波的强度与粒子在该处附近出现的概率成正比 — **几率波**

$$|\Psi(r, t)|^2 = \Psi\Psi^* = \Psi_0^2 \quad \longrightarrow \quad \text{Probability density}$$

Meaning: at some instant, the **probability** of finding the particle within a **unit volume** at that location.

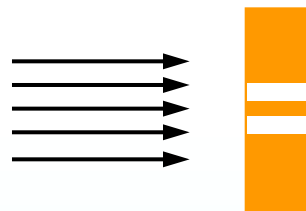
表示: t 时刻粒子出现在空间某点 r 处单位体积内的概率。

$$dW = |\Psi(r, t)|^2 dV$$

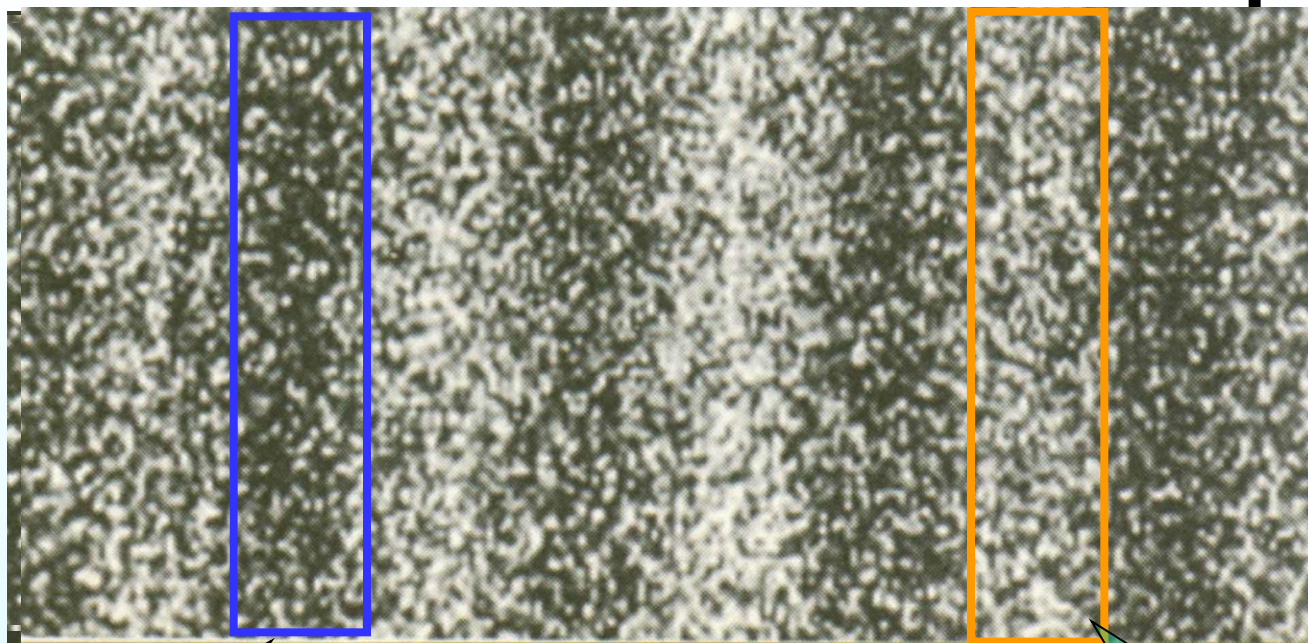
Meaning : at some instant, the **probability** of finding the particle within the **volume dV** about the given position

表示: t 时刻粒子出现在空间某点 r 附近体积元 dV 内的概率。

单个粒子在哪一处出现是偶然事件；
大量粒子的分布有确定的统计规律。



电子双缝干涉图样



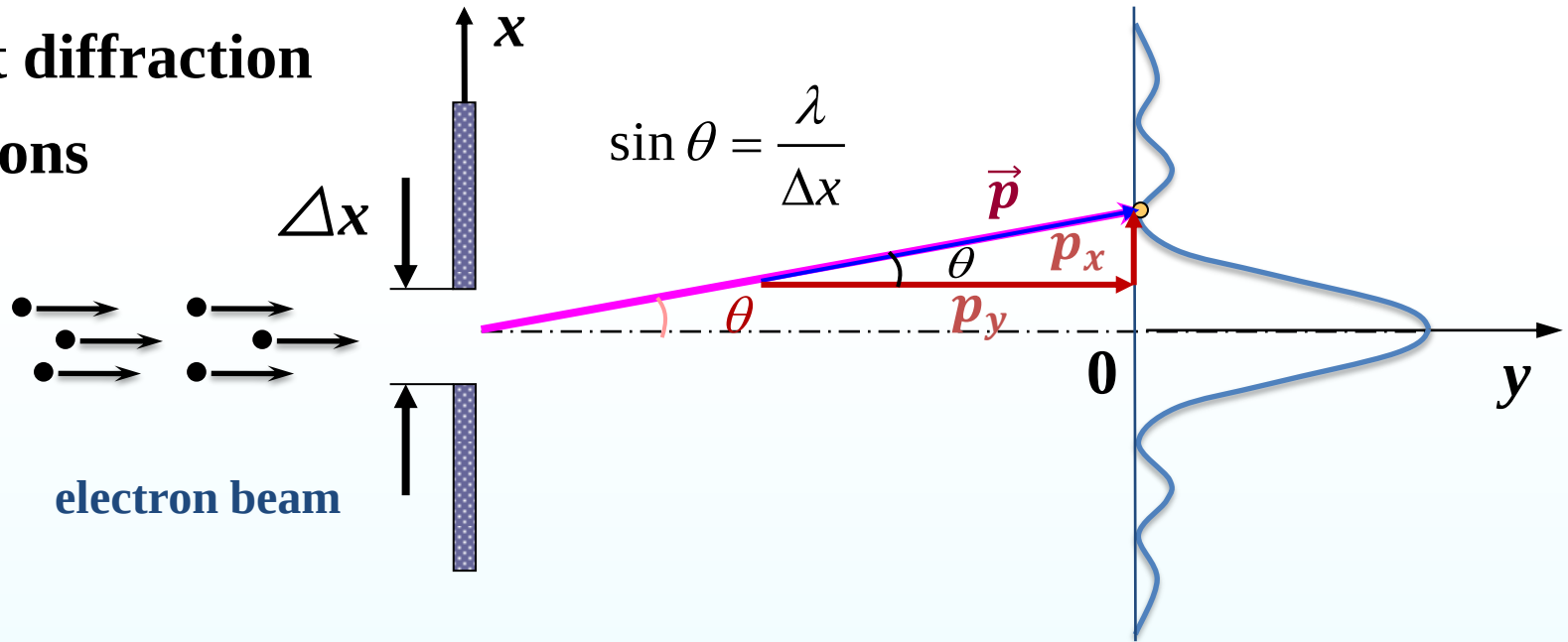
出现概率小

电子数 $N=70000$

出现概率大

23.2 The Heisenberg Uncertainty principle

one-slit diffraction
of electrons



position uncertainty (坐标的不确定量) : Δx

momentum uncertainty (动量的不确定量): Δp_x

first order
diffraction

$$\Delta x \sin \theta = \lambda \rightarrow \Delta x = \frac{\lambda}{\sin \theta}$$

$$p_x = \Delta p_x = p \sin \theta$$
$$p = h/\lambda$$

$$\Delta p_x = p_x = \frac{h}{\lambda} \sin \theta$$

$$\Delta x \cdot \Delta p_x = h$$

Involving other orders

$$\Delta x \cdot \Delta P_x \geq h$$

More careful calculation

$$\left. \begin{aligned} \Delta x \cdot \Delta p_x &\geq \hbar \\ \Delta y \cdot \Delta p_y &\geq \hbar \\ \Delta z \cdot \Delta p_z &\geq \hbar \end{aligned} \right\}$$

Heisenberg
uncertainty
principle



Werner Karl
Heisenberg,
1901-1976,
1932 “Noble”

We can not measure both the position and momentum of an object precisely **at the same time**. The more accurately we try to measure the position, the greater will be the uncertainty in momentum.

- **Uncertainty principle relates energy and time**

The object to be detected has an **uncertainty** in **position**

$$\Delta x \approx \lambda$$

The photon used to detect it travels with speed c and take a **time** to pass through the **distance of uncertainty**

$$\Delta t = \frac{\lambda}{c}$$

The photon can **transfer** some or all of its **energy** to the given object, the **uncertainty** in **energy** of the objects as a result is

$$\Delta E = hf = \frac{hc}{\lambda}$$

The **product** of these two uncertainties is

$$\Delta E \Delta t \approx h$$

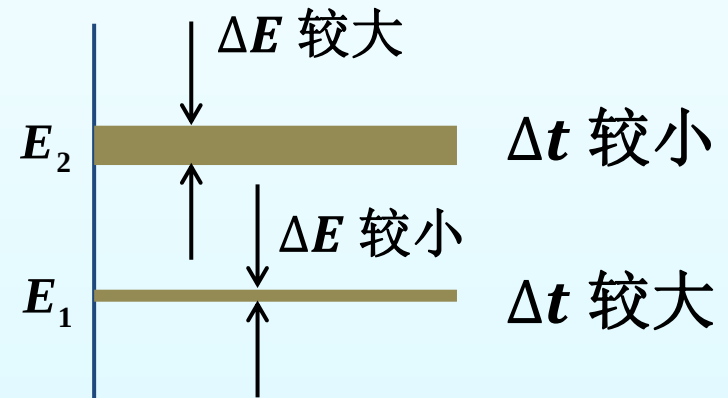
More careful calculations gives

$$\Delta E \Delta t \geq \hbar$$

The **energy** of an object can be **uncertain**, or may even be **nonconserved**, by an amount ΔE for a time Δt .

The **longer** the lifetime of a state, the **smaller** is its spread in energy.

激发态	{	平均寿命 $\Delta t \sim 10^{-8}\text{s}$
		能级宽度 $\Delta E \geq \frac{\hbar}{\Delta t} \sim 10^{-8}\text{ eV}$
基态	{	平均寿命 $\Delta t \rightarrow \infty$
		能级宽度 $\Delta E \rightarrow 0$



Discussion:

1. The uncertainty is **inherent** in nature and it is the result of two factors: the **wave-particle duality** and the unavoidable interaction between the thing observed and the observing instrument.
2. Any moving particle has **no** certain **obits**. What we can get is the probability of its position, momentum at next instant.
3. For macro objects, **h** is **so small** and the uncertainties are unusually negligible.

Example For a tiny dust, macroscopic particle $m = 10^{-10}\text{kg}$, its position uncertainty could be $\Delta x = 10^{-10}\text{m}$.

Determine $\Delta v_x = ?$

Solution:

$$\Delta x \cdot \Delta P_x \geq \hbar$$

$$\Delta x \cdot m \Delta v_x \geq \hbar$$

$$\Delta v_x \geq \frac{\hbar}{m \Delta x} = 1.06 \times 10^{-18} \text{ m / s}$$

The uncertainties are usually **negligible** on the macroscopic level.

Example Position uncertainty of electron

An **electron** moves in a straight line with a **constant speed** $v = 1.10 \times 10^6$ m/s which has been measured to a **precision** of 0.10 %. What is the maximum **precision** with which its **position** could be simultaneously measured?

Solution: $p = mv = 1.00 \times 10^{-24}$ kg · m/s

$$\Delta p = 1.0 \times 10^{-27} \text{ kg} \cdot \text{m/s}$$

$$\Delta x \geq \frac{\hbar}{\Delta p} = \frac{1.06 \times 10^{-34}}{1.0 \times 10^{-27}} = 1.1 \times 10^{-7} \text{ m}$$

This is about **1000 times** the diameter of an atom.

23.3 The Schrödinger equation

Schrödinger principal contribution was to formulate a differential equation that was satisfied by the amplitude of the de Broglie wave.

1. The time-dependent Schrödinger equation

(1) **A free particle m** (with no force on it)
in one dimension (moving along the x axis)

Wave function

$$\Psi(x, t) = \Psi_o e^{-\frac{i}{\hbar}(Et - Px)}$$

Wave-particle duality

$$f = \frac{E}{h} \quad \lambda = \frac{h}{p}$$



Erwin Schrodinger
1933 “Noble”

Take the **first derivative** with respect to **t**

$$\frac{\partial \Psi}{\partial t} = -\frac{i}{\hbar} E \Psi_0 e^{-\frac{i}{\hbar}(Et - Px)} = -\frac{i}{\hbar} E \Psi$$

Take the **second derivative** with respect to **x**

$$\frac{\partial^2 \Psi}{\partial x^2} = -\frac{P^2}{\hbar^2} \Psi_0 e^{-\frac{i}{\hbar}(Et - Px)} = -\frac{P^2}{\hbar^2} \Psi$$

a free particle $v \ll c$ $E = K$ $p^2 = 2mE$

based on the **energy conservation**

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} = i\hbar \frac{\partial \Psi}{\partial t}$$

**One –dimensional time– dependent Schr ödinger equation
for a free particle**

(2) One-dimensional time-dependent Schrödinger equation for a particle with potential energy $U(\mathbf{x}, t)$

$$\begin{aligned} E = K + U = \frac{P^2}{2m} + U & \left. \begin{aligned} \frac{\partial \Psi}{\partial t} &= -\frac{i}{\hbar} E \Psi \\ \frac{\partial^2 \Psi}{\partial x^2} &= -\frac{P^2}{\hbar^2} \Psi \end{aligned} \right\} \frac{\partial \Psi}{\partial t} = -\frac{i}{\hbar} \left[\frac{P^2}{2m} + U(x, t) \right] \Psi \\ -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} &= \frac{P^2}{2m} \Psi \end{aligned}$$
$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + U(x, t) \Psi = i\hbar \frac{\partial \Psi}{\partial t}$$

一维运动粒子含时薛定谔方程

(3) three –dimensional time-dependent Schrödinger equation for a particle with potential energy $U(x,y,z,t)$

$$-\frac{\hbar^2}{2m}\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right]\Psi + U(x,y,z,t)\Psi = i\hbar \frac{\partial \Psi}{\partial t}$$

Laplace operator $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

General form

$$-\frac{\hbar^2}{2m}\nabla^2\Psi(\vec{r},t) + U(\vec{r},t)\Psi(\vec{r},t) = i\hbar \frac{\partial \Psi(\vec{r},t)}{\partial t}$$

薛定谔方程是**非相对论量子力学**的基本动力学方程，其地位与经典力学中的牛顿方程相同。

2.The time-independent Schrödinger equation

The function of potential energy is **not changeable** with time t ,
 $U(x, y, z, t) \rightarrow U(\vec{r})$ - **time-independent** state(定态) .

$$\begin{aligned}\Psi(x, t) &= \Psi_0 e^{-i(Et - \vec{p} \cdot \vec{r}) / \hbar} \\ &= \Psi_0 e^{i\vec{p} \cdot \vec{r} / \hbar} e^{-iEt / \hbar} = \psi(\vec{r}) f(t)\end{aligned}$$

Substitute Ψ into S equation:

$$\begin{aligned}-\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r}) f(t) + U(\vec{r}) \psi(\vec{r}) f(t) &= i\hbar \frac{\partial \psi(\vec{r}) f(t)}{\partial t} \\ \frac{1}{\psi(\vec{r}) f(t)} \rightarrow -\frac{\hbar^2}{2m} \frac{1}{\psi(\vec{r})} \nabla^2 \psi(\vec{r}) + U(\vec{r}) &= i\hbar \frac{1}{f(t)} \frac{\partial f(t)}{\partial t} = E\end{aligned}$$

$$-\frac{\hbar^2}{2m} \frac{1}{\psi(\vec{r})} \nabla^2 \psi(\vec{r}) + U(\vec{r}) = i\hbar \frac{1}{f(t)} \frac{\partial f(t)}{\partial t} = E$$

$$i\hbar \frac{\partial}{\partial t} f(t) = E f(t) \quad \longrightarrow \quad f(t) = C e^{-\frac{i}{\hbar} E t}$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r}) + U \psi(\vec{r}) = E \psi(\vec{r})$$

$$\longrightarrow \frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r}) + (E - U) \psi(\vec{r}) = 0$$

time-independent Schrödinger in one dimension
(一维运动粒子定态薛定谔方程)

$$\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} + (E - U) \psi(x) = 0$$

$$\frac{d^2 \psi(x)}{dx^2} + \frac{2m}{\hbar^2} (E - U) \psi(x) = 0$$

3. Some requirements imposed on wave function

(1) It should be a **continuous** function

Since the probability should be continuous from point to point and not to take discontinuous jumps.

波函数：单值、连续、有限

(2) It should be **normalized** (归一化)

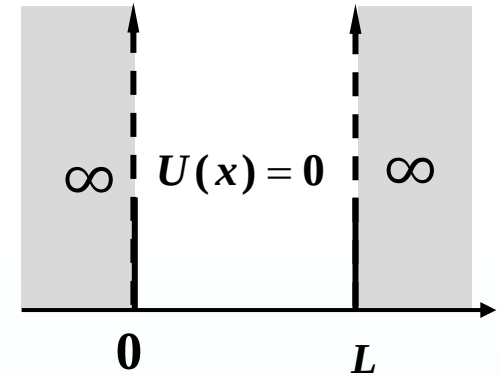
For a single particle, the probability of finding the particle summed over all space must be exactly 1.

$$\int_{all\ space} |\psi|^2 dV = 1$$

23.5 Particle in an infinitely deep square potential well

Potential energy $U(x)$ as

$$U(x) = \begin{cases} 0, & 0 < x < L \\ \infty, & x \leq 0, x \geq L \end{cases}$$



When $x \leq 0, x \geq L \rightarrow \psi(x) = 0$

Electrons can **not** jump out of the well.

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \quad \rightarrow \quad \frac{d^2\psi}{dx^2} + k^2\psi = 0$$

Denote: $k = \sqrt{\frac{2mE}{\hbar^2}}$

Angular wave number k

General solution: $\psi(x) = A \sin kx + B \cos kx$

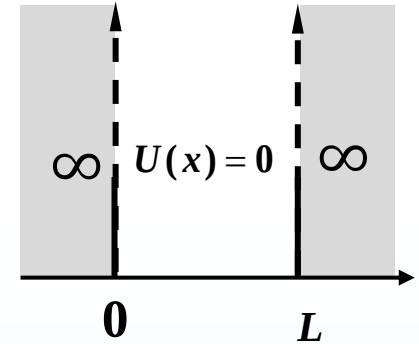
Determine A, B, k (ψ is continuous) $\psi(x) = A \sin kx + B \cos kx$

Boundary conditions

$$\psi(0) = \psi(L) = 0$$

$$\psi(0) = B \cos kx = 0 \rightarrow B = 0$$

$$\psi(L) = A \cos kL = 0 \rightarrow kL = n\pi$$



$$k = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots$$

Quantum number

$$k = \sqrt{\frac{8\pi^2 m E}{h^2}}$$

$$E_n = n^2 \frac{h^2}{8mL^2}$$

$$E_n = n^2 E_1$$

$$\psi(x) = A \sin kx \quad k = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots$$

$$\psi(x) = A \sin \frac{n\pi}{L} x$$

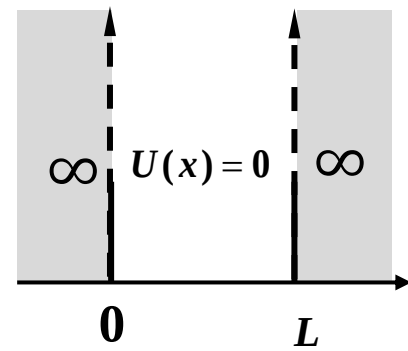
Normalization (归一化): $\int_{-\infty}^{\infty} |\psi|^2 dx = \int_0^a \psi \psi^* dx = 1$

$$A^2 \int_0^a \sin^2 \frac{n\pi}{L} x dx = 1 \quad A = \sqrt{\frac{2}{L}}$$

$$\therefore \psi(x) = \sqrt{\frac{2}{L}} \sin kx \quad k = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots$$

◆ **wave equation** $\frac{d^2\psi}{dx^2} + \frac{8\pi^2 mE}{h^2}\psi = 0$

◆ **wave function**



$$\psi(x) = \begin{cases} 0, & (x \leq 0, x \geq L) \\ \sqrt{\frac{2}{L}} \sin \frac{n\pi}{L} x, & (0 < x < L) \end{cases}$$

◆ **probability density** $|\psi(x)|^2 = \frac{2}{L} \sin^2 \frac{n\pi}{L} x$

◆ **energy** $E_n = n^2 \frac{h^2}{8mL^2}$

$$E_n = n^2 E_1$$

Discussion:

1. Energy is quantized

Energy

$$E_n = n^2 \frac{h^2}{8mL^2}$$

ground energy

$$E_1 = \frac{h^2}{8mL^2}, \quad (n = 1)$$

excited energy

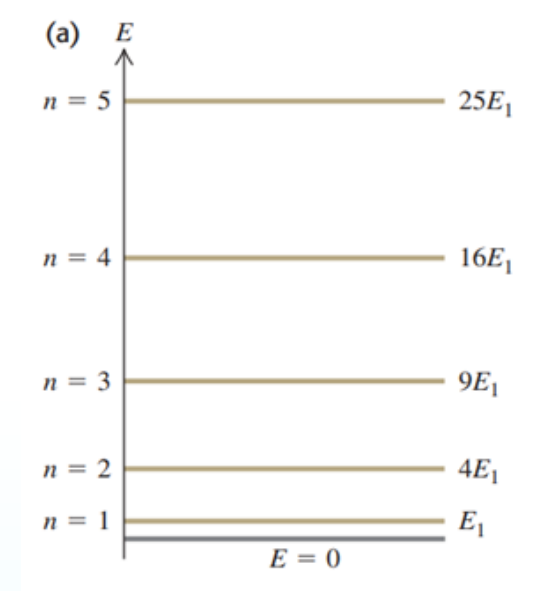
$$E_n = n^2 \frac{h^2}{8mL^2} = n^2 E_1, \quad (n = 2, 3, \dots)$$

$$(a) \frac{\Delta E_n}{E_n} = \frac{(n+1)^2 - n^2}{n^2} = \frac{(2n+1)}{n^2}$$

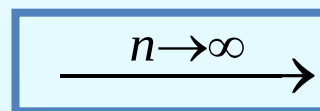
$$\text{When } n \rightarrow \infty, \quad \frac{\Delta E_n}{E_n} \rightarrow 0$$

$$(b) L = 10^{-10} \text{ m}, \quad \Delta E_n \sim 3.76(2n+1) \text{ eV}$$

$$L = 1 \text{ cm}, \quad \Delta E_n \sim 3.76 \times 10^{-16} (2n+1) \text{ eV}$$



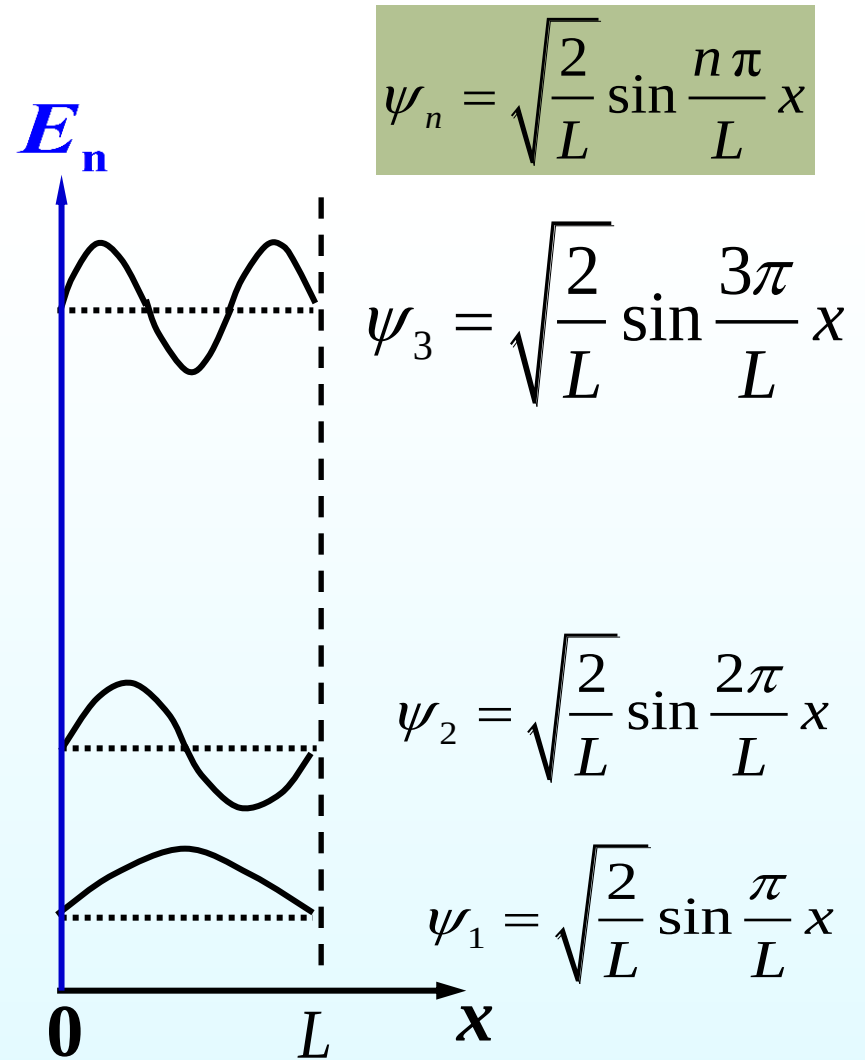
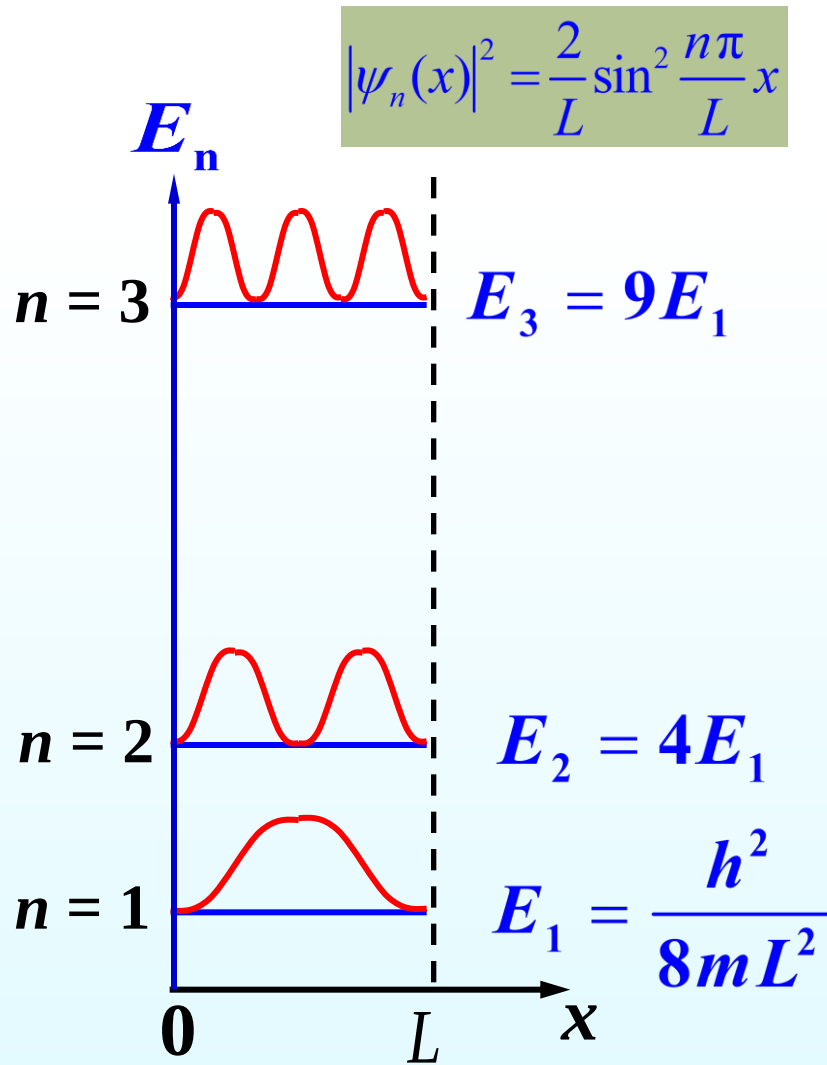
Quantum mechanics



classical mechanics

Correspondence principle

2. Probability function and probability density



$$|\psi_n(x)|^2 = \frac{2}{L} \sin^2 \frac{n\pi}{L} x$$

maximum probability density

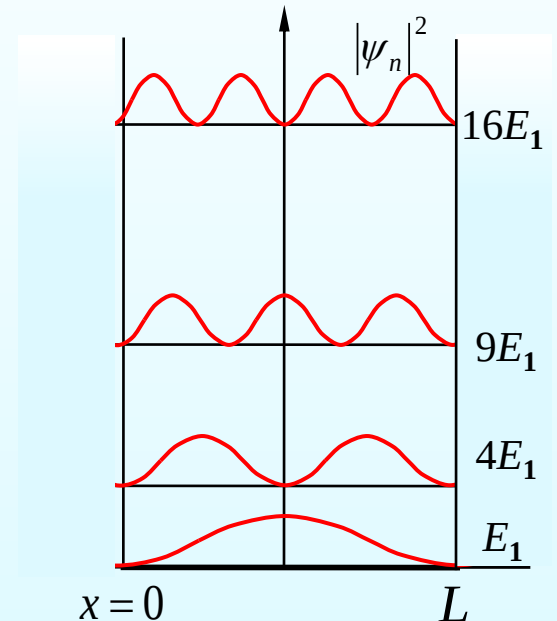
$$\frac{n\pi}{L} x = k\pi + \frac{\pi}{2} \rightarrow x = \frac{(2k+1)L}{2n} \quad (k = 0, 1, 2, \dots)$$

The number of positions for maxima : n

minimum probability density

$$\frac{n\pi}{L} x = k\pi \rightarrow x = \frac{kL}{n} \quad (k = 0, 1, 2, \dots)$$

The number of positions for minima : $n+1$

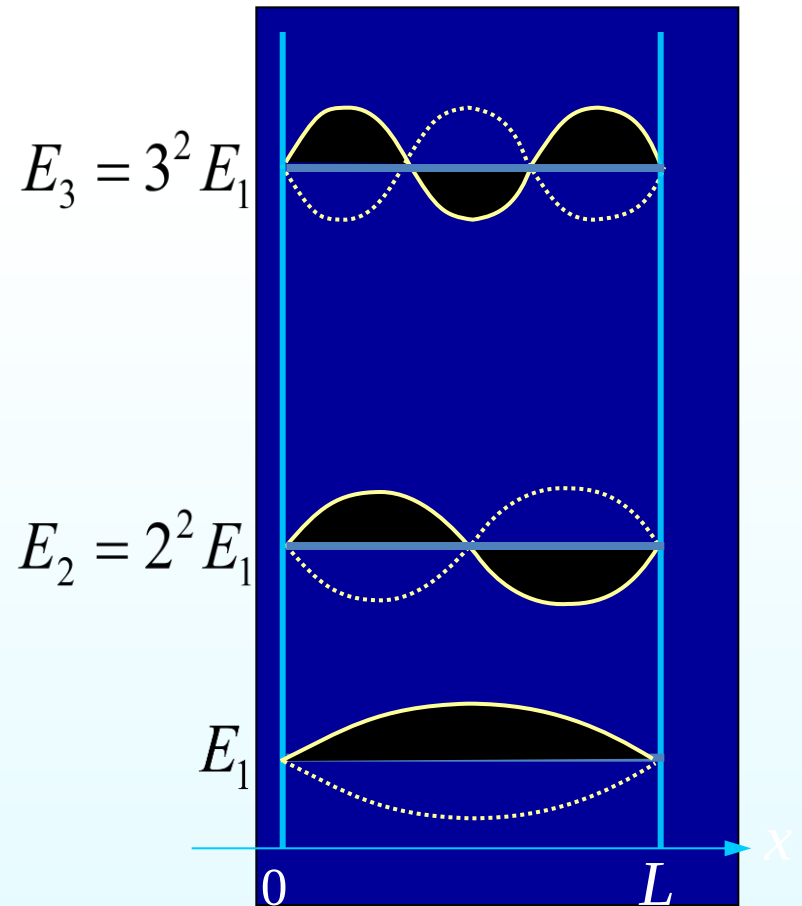


3. The property of standing wave

$$L = n \frac{\lambda}{2}$$

$$p_n = \frac{h}{\lambda} = \frac{nh}{2L}$$

$$E_n = \frac{p_n^2}{2m} = \frac{n^2 h^2}{8mL^2} = n^2 E_1$$

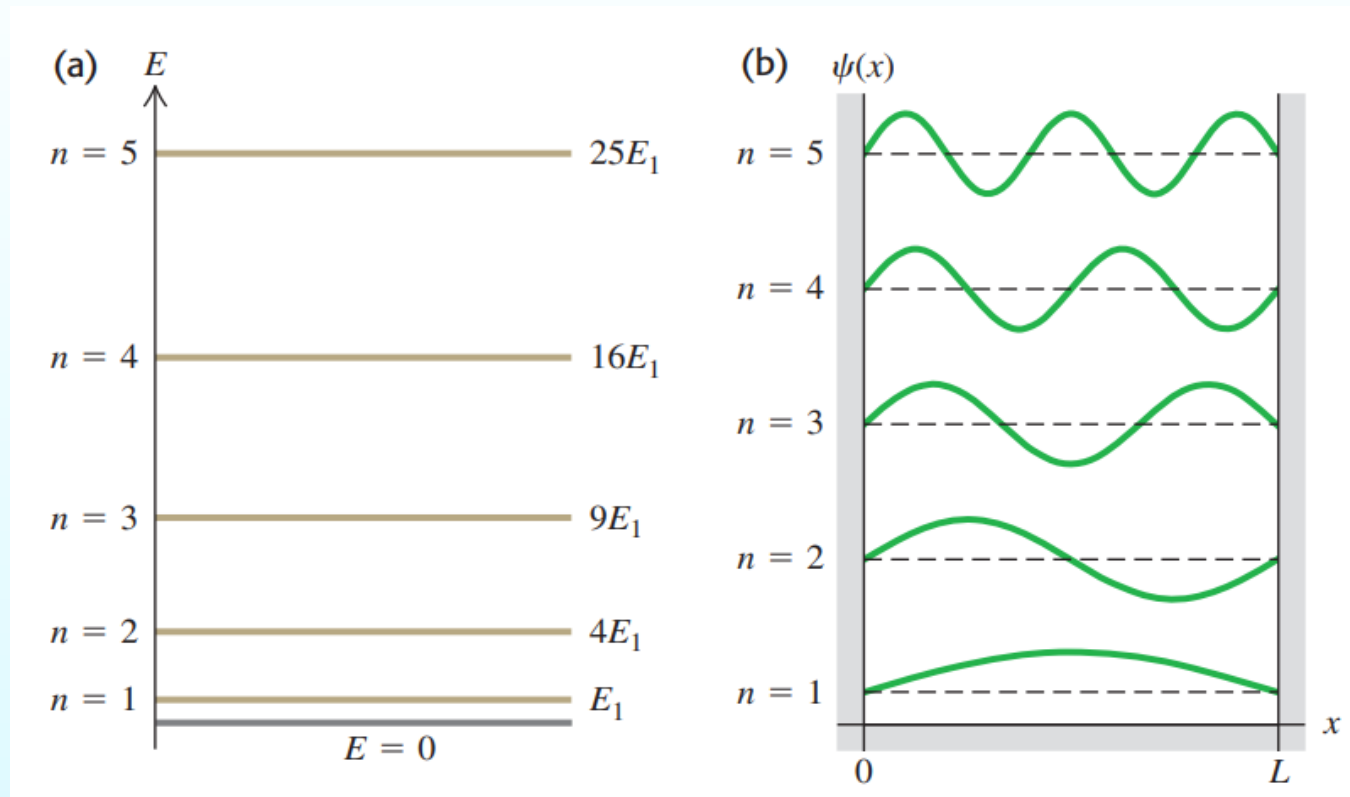


Wave function

A particle is confined to a one-dimensional trap by infinite potential energy walls. Of the following states, designed by the quantum number n , for which one is the probability density *greatest* near the *center* of the well?



- A. $n = 2$
- B. $n = 3$
- C. $n = 4$
- D. $n = 5$
- E. $n = 6$



Example A **ground-state** electron is trapped in the one-dimensional infinite potential well, with width $L=100\text{pm}$.
 (a) What is the **probability** that the electron can be detected in the left one-third of the well ($0 \sim L/3$). (b) What is the **probability** that the electron can be detected in the middle one-third of the well? ($L/3 \sim 2L/3$)

Solution:
$$|\psi_n(x)|^2 = \frac{2}{L} \sin^2 \frac{n\pi x}{L} \quad (n=1)$$

$$(a)P = \int_0^{\frac{L}{3}} |\psi_n(x)|^2 dx = \int_0^{\frac{L}{3}} \frac{2}{L} \sin^2 \frac{\pi x}{L} dx = \frac{2}{L} \int_0^{\frac{L}{3}} \left(\frac{1 - \cos 2\pi x / L}{2} \right) dx$$

$$= \frac{1}{3} - \frac{\sqrt{3}}{4\pi} = 0.20$$

$$(b)P = 1 - 0.20 - 0.20 = 0.60$$

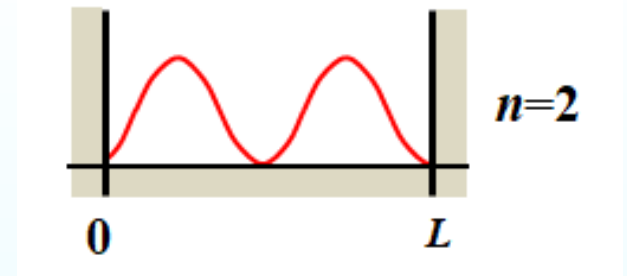


$n=1$

Example An electron moves in the one dimensional infinite potential well width L .

Determine (a) Where are the **positions of maximum probability density** when the electron is in the **first excited state**. (b) the **probability** of finding the electron with the range, **0 to $L/4$** of the well.

Solution: $|\psi_n(x)|^2 = \frac{2}{L} \sin^2 \frac{n\pi x}{L}$



$$(a) \frac{n\pi}{L} x = k\pi + \frac{\pi}{2} \rightarrow x = \frac{(2k+1)L}{2n} \quad (k = 0, 1, 2, \dots)$$

$$n = 2, \quad x = \frac{L}{4}, \quad \frac{3L}{4}$$

$$(b) P = \int_0^{\frac{L}{4}} |\psi_n(x)|^2 dx = \int_0^{\frac{L}{4}} \frac{2}{L} \sin^2 \frac{2\pi x}{L} dx = \frac{1}{4} - \frac{1}{4\pi} \sin \pi = \frac{1}{4}$$

Example Confined bacterium. A tiny bacterium with a mass of about 10^{-14}kg is confined between two rigid walls 0.1mm apart. (a) Estimate its **minimum speed**. (b) If, instead, its speed is about 1mm in 100s , estimate the **quantum number** of its state.

Solution: (a) The minimum speed in the ground state $n=1$

$$v = \sqrt{\frac{2E}{m}} = \sqrt{\frac{h^2}{4m^2L^2}} = \frac{h}{2mL} = \frac{6.6 \times 10^{-34}}{2 \times 10^{-14} \times 10^{-4}} = 3 \times 10^{-16} \text{ m/s}$$

$$\text{(b)} \quad E = \frac{1}{2}mv^2 = \frac{1}{2} \times 10^{-14} \times (10^{-5})^2 = 5 \times 10^{-25} \text{ J}$$

$$E = \frac{n^2 h^2}{8mL^2} \Rightarrow n = \sqrt{E \left(\frac{8mL^2}{h^2} \right)} \approx 3 \times 10^{10}$$

例 某粒子运动时的波函数为

$$\psi(x) = \begin{cases} \sqrt{\frac{2}{L}} \cos \frac{3\pi}{L} x, & |x| < \frac{L}{2} \\ 0, & |x| > \frac{L}{2} \end{cases}$$

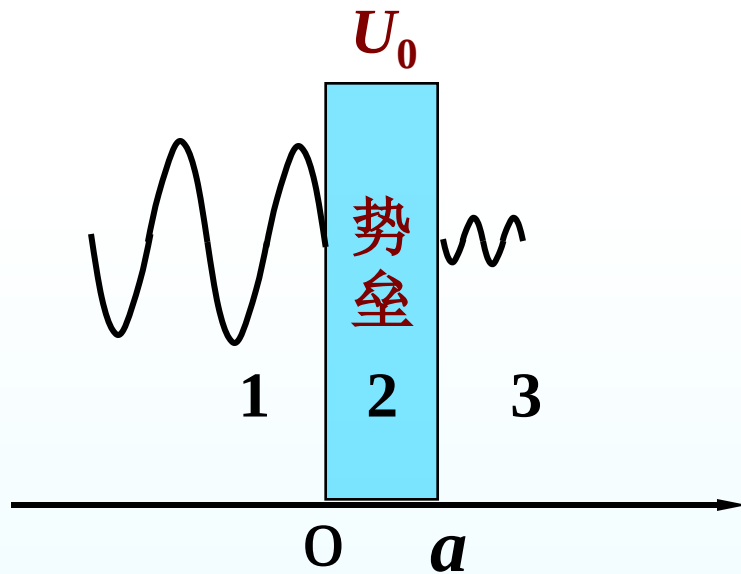
试求：(1) 概率密度的表达式；(2) 粒子出现概率最大的各个位置.

解： (1) $|\psi_n(x)|^2 = \frac{2}{L} \cos^2 \frac{3\pi x}{L}, \quad -\frac{L}{2} < x < \frac{L}{2}$

$$(2) \cos^2 \frac{3\pi x}{L} = 1 \quad \frac{3\pi x}{L} = k\pi \rightarrow x = \frac{kL}{3} \quad (k = -1, 0, 1)$$

$$x = -\frac{L}{3}, 0, \frac{L}{3}$$

Tunneling through a barrier



经典理论

- 1. $E > U_0$ 的粒子，
越过势垒。
- 2. $E < U_0$ 的粒子，
不能越过势垒。

量子理论

- 1. $E > U_0$ 的粒子，也存在被弹回的概率
—— 反射波
- 2. $E < U_0$ 的粒子，也可能越过势垒到达3区
—— 隧道效应

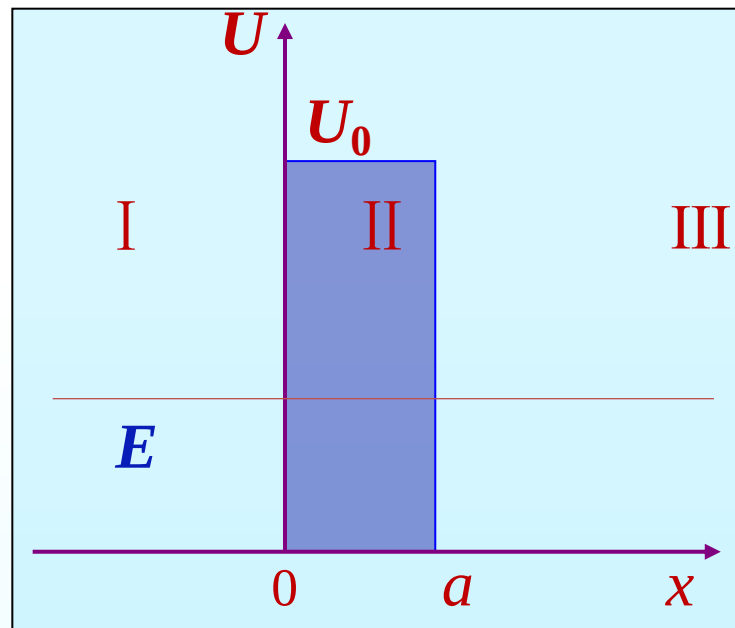
势能函数

$$\begin{cases} U(x) = 0 & x \leq 0 \text{ (I区)} \\ U(x) = U_0 & 0 \leq x \leq a \text{ (II区)} \\ U(x) = 0 & x \geq a \text{ (III区)} \end{cases}$$

$$\frac{d^2\Psi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - U)\Psi(x) = 0$$

波函数

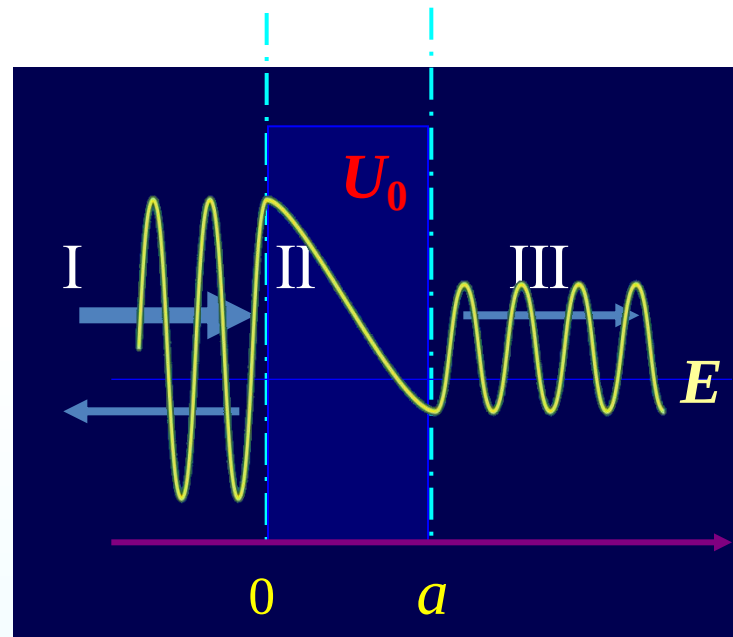
$$\begin{cases} \Psi_1(x) = A_1 e^{ik_1 x} + B_1 e^{-ik_1 x} & \text{(I区)} \\ \Psi_2(x) = A_2 e^{k_2 x} + B_2 e^{-k_2 x} & \text{(II区)} \\ \Psi_3(x) = A_3 e^{ik_1 x} & \text{(III区)} \end{cases}$$
$$\begin{aligned} k_1^2 &= \frac{2mE}{\hbar^2} \\ k_2^2 &= \frac{2m(U_0 - E)}{\hbar^2} \\ k_1^2 &= \frac{2mE}{\hbar^2} \end{aligned}$$



利用波函数在边界处连续
(包括导数) 特点, 可求解式
中各系数之间关系

可求出透射系数为

$$T = \frac{A_3^2}{A_1^2} \propto e^{-\frac{2a}{\hbar} \sqrt{2m(U_0 - E)}} \neq 0$$



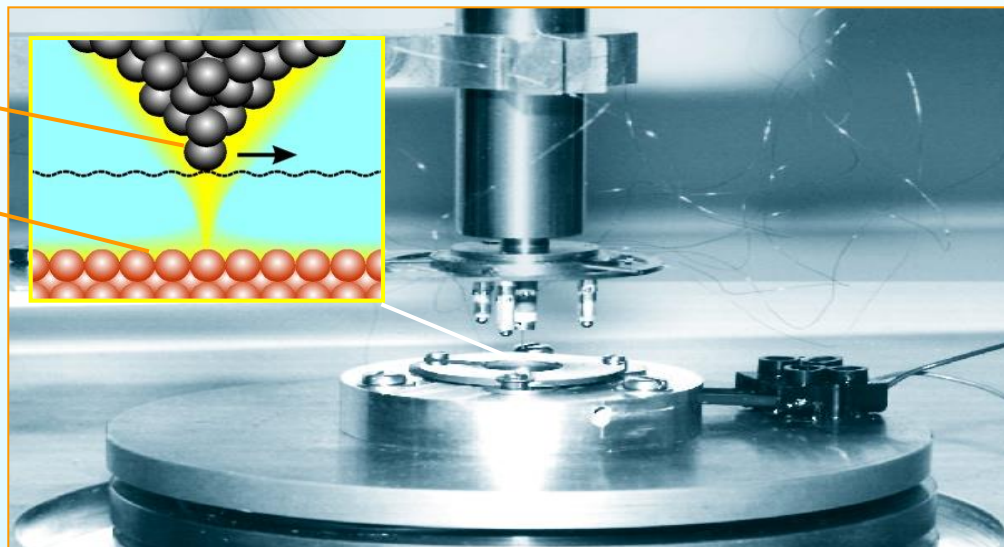
可见, 粒子的贯穿系数与势垒的宽度及高度有关

应用

1981年宾尼希和罗雷尔利用电子的隧道效应制成了扫描隧道显微镜(STM). 它利用针尖扫描样品表面, 通过隧道电流获得样品表面的图像. 可观测固体表面原子排列的状况, 还可进行单原子操作, 实现了初步的原子结构制造技术.

1986年宾尼希又研制了原子力显微镜.

探针
样品表面



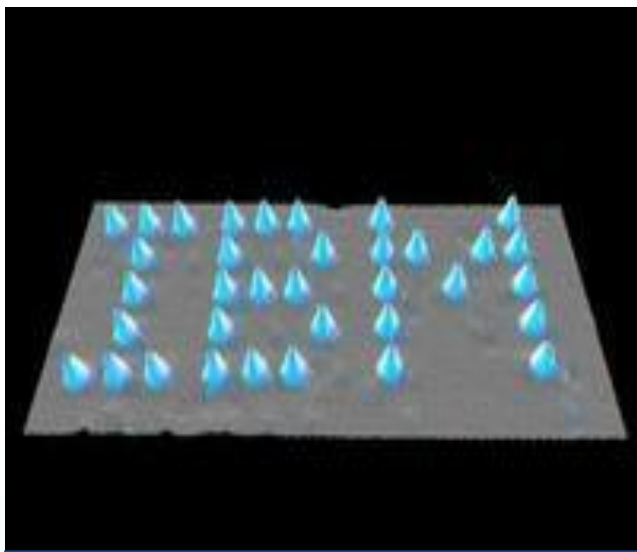
G. Binnig

1986 Nobel
Prize in Physics

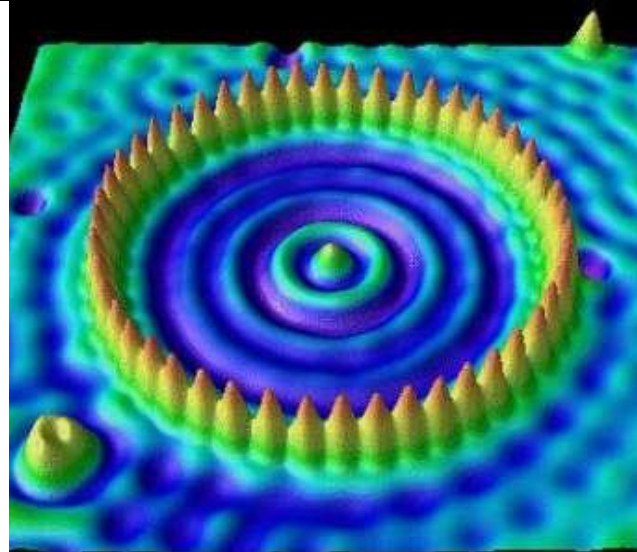


H. Rohrer

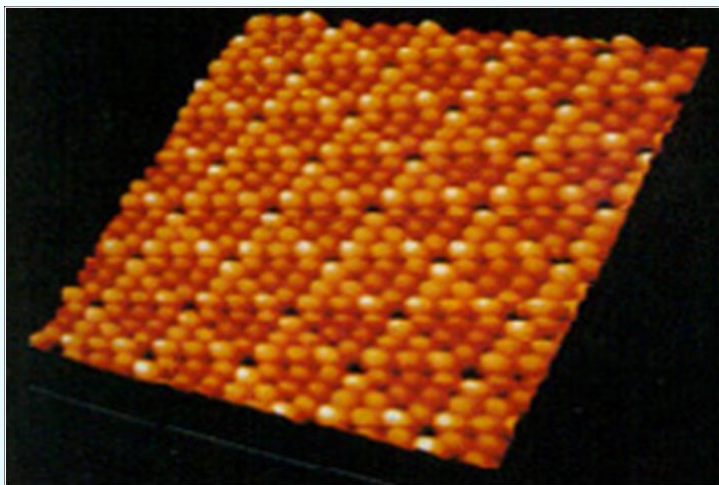
1986 Nobel
Prize in Physics



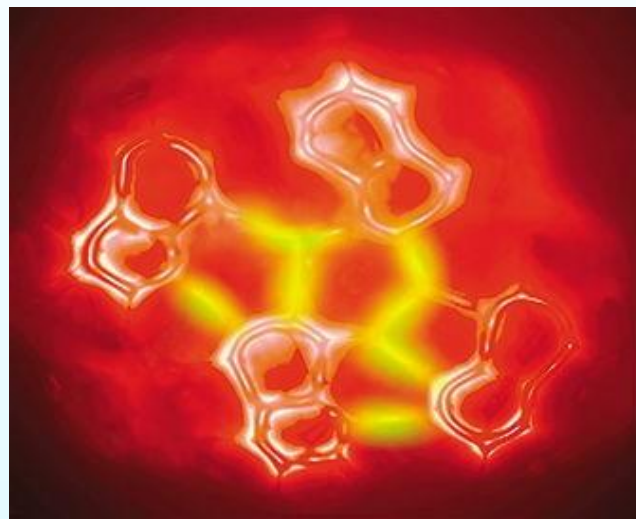
原子商标



量子围栏照片



硅表面硅原子的排列



原子力显微镜观察
分子间的氢键

Key words And Terms

quantum states

量子态

principal quantum number

主量子数

orbital quantum number

(轨道) 角量子数

magnetic quantum number

(轨道) 磁量子数

spin quantum number

自旋量子数

spin magnetic number

自旋磁量子数

periodic table

元素周期表

atomic number

原子序数

electronic configuration of the ground state

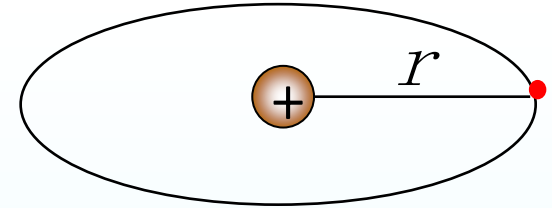
基态原子组态

23.5 Schrödinger equation for hydrogen atom

1. Schrödinger equation for hydrogen atoms

Potential energy

$$U = -\frac{e^2}{4\pi\epsilon_0 r}$$

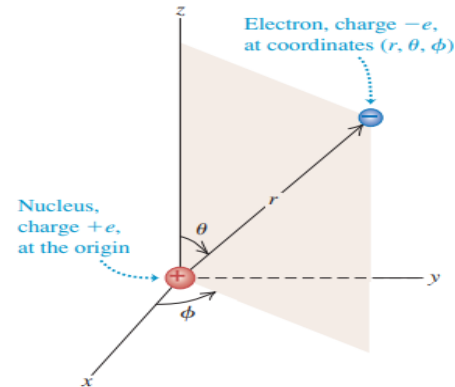


time-independent Schrödinger equation

$$\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r}) + (E - U) \psi(\vec{r}) = 0$$

Spherical coordinates

$$\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases}$$



$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{8\pi^2 m}{h^2} \left(E + \frac{e^2}{4\pi\epsilon_0 r} \right) \psi = 0$$

separate variables

$$\psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi)$$

$$\left\{ \begin{array}{l} \frac{d^2 \Phi}{d\phi^2} + m_l^2 \Phi = 0 \\ \frac{m_l^2}{\sin^2 \theta} - \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = l(l+1) \\ \frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{8\pi^2 m r^2}{h^2} \left(E + \frac{e^2}{4\pi\epsilon_0 r} \right) = l(l+1) \end{array} \right.$$

2. Quantized condition and quantum numbers

(1) The energy and principal quantum number n (主量子数)

$$E_n = -\frac{me^4}{2\hbar^2(4\pi\epsilon_0)^2} \frac{1}{n^2} = -\frac{me^4}{8\hbar^2\epsilon_0^2} \frac{1}{n^2}$$

$$E_n = -13.6 \frac{1}{n^2} \text{ (eV)} \quad n = 1, 2, 3, \dots,$$

a. n determines the energy of atom and energy is quantized

b. when $n \rightarrow \infty$, $E_n \rightarrow$ continuous

(2) Orbital quantum number l (轨道)角量子数

l determines orbital angular momentum L for that the electrons orbit about the nucleus. In atoms with two or more electrons, the energy does depend on l as well as n .

$$L = \sqrt{l(l+1)} \frac{h}{2\pi} = \sqrt{l(l+1)} \hbar$$

$$l = 0, 1, 2, \dots, (n-1)$$

For example: $l = 0, 1$

$$L = 0, \quad \sqrt{2}\hbar$$

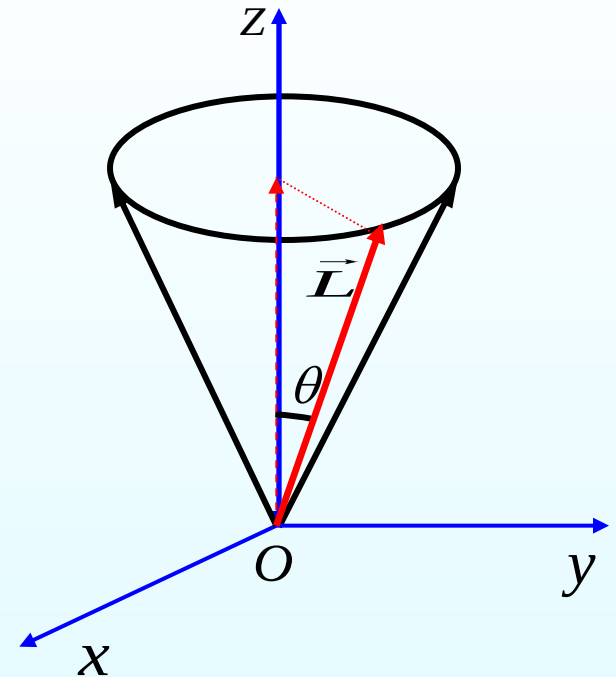
(3) Magnetic quantum number m_l 磁量子数

m_l is related to the **direction** of the electron's orbital angular momentum $\vec{L}$. The component (in a magnetic field) of angular momentum along some fixed direction, usually chosen as the Z direction.

$$L_z = m_l \hbar$$

$$m_l = 0, \pm 1, \pm 2, \dots, \pm l$$

Number: $2l+1$



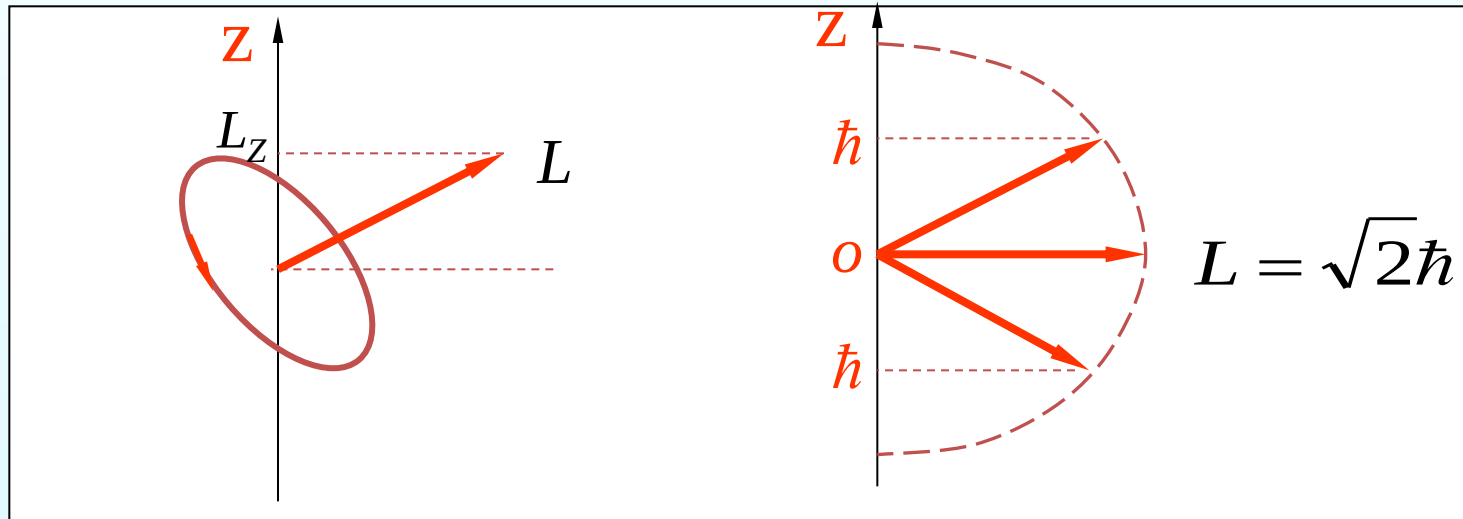
This limitation on the direction of angular momentum is often called **space quantization** —角动量空间量子化

For example: $l = 1$

$$L = \sqrt{l(l+1)} \frac{h}{2\pi} = \sqrt{2} \frac{h}{2\pi} = \sqrt{2}\hbar$$

then , $m_l = 0, \pm 1,$

$$L_z = 0, \frac{h}{2\pi}, -\frac{h}{2\pi}$$



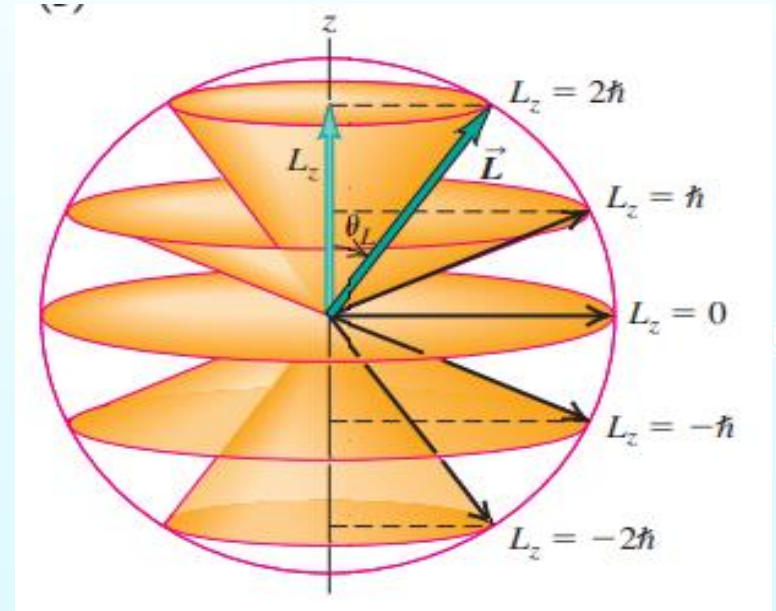
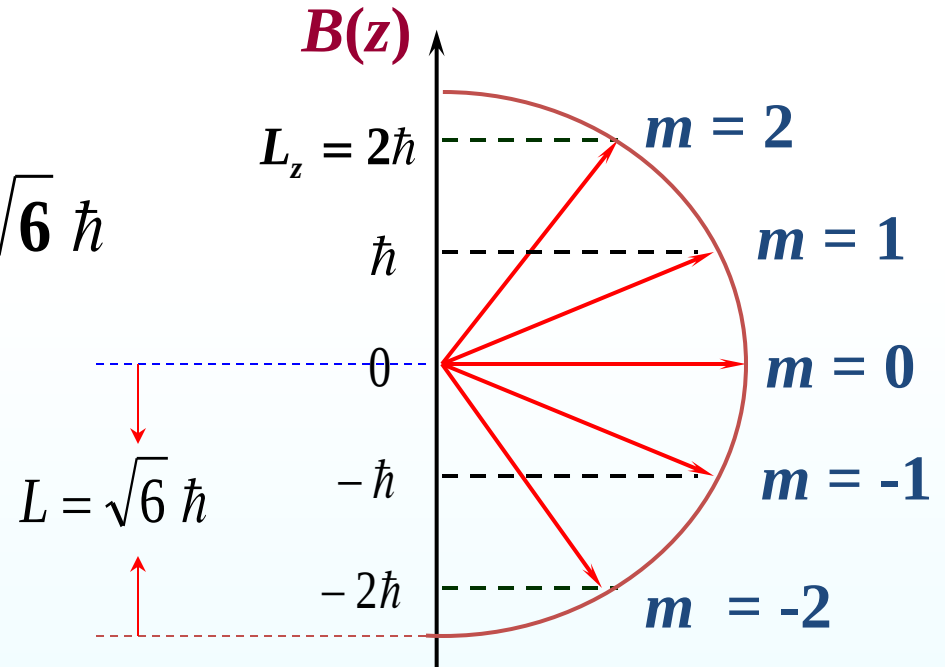
$$l = 2$$

$$L = \sqrt{l(l+1)} \hbar = \sqrt{2(2+1)} \hbar = \sqrt{6} \hbar$$

$$L_z = m_l \hbar$$

$$m_l = 0, \pm 1, \pm 2, \dots, \pm l$$

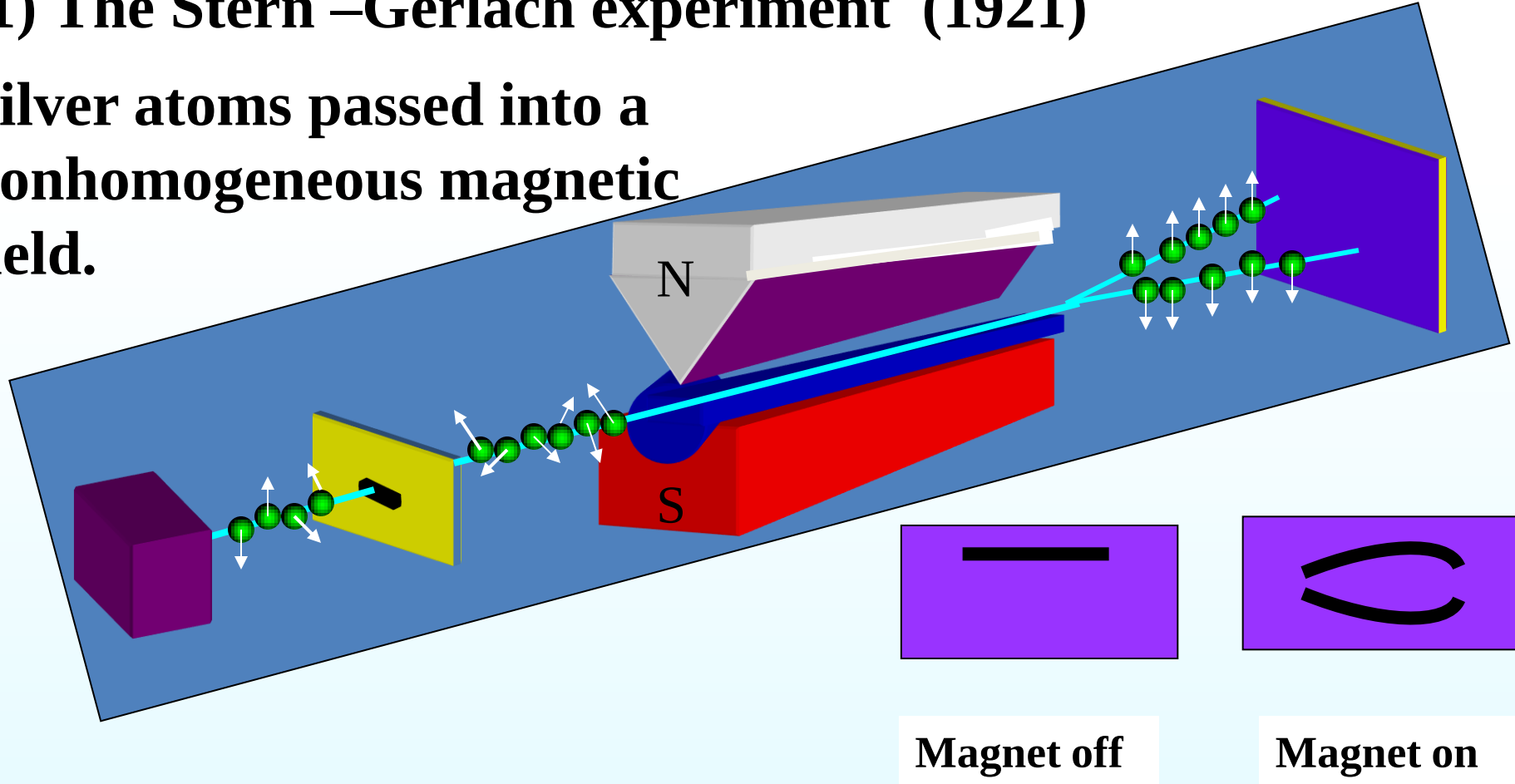
$$L_z = 0, \pm \hbar, \pm 2\hbar$$



3. Spin of electrons

(1) The Stern –Gerlach experiment (1921)

Silver atoms passed into a nonhomogeneous magnetic field.



If the lines were due to orbital angular momentum, there should have been an *odd* number, corresponding to the possible values of m_l .

(2) Spin of electrons

Spin angular momentum

$$S = \sqrt{s(s+1)}\hbar \Rightarrow S = \frac{\sqrt{3}}{2}\hbar$$

$$s = \frac{1}{2}$$

s : spin quantum number 自旋量子数

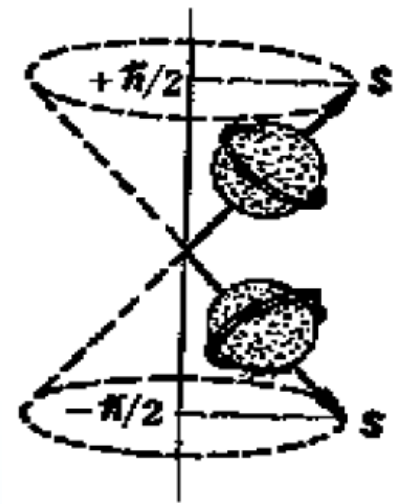
$$S_z = m_s \hbar$$

$$m_s = \pm \frac{1}{2}$$

m_s : spin magnetic quantum number

自旋磁量子数

+ spin up, - spin down

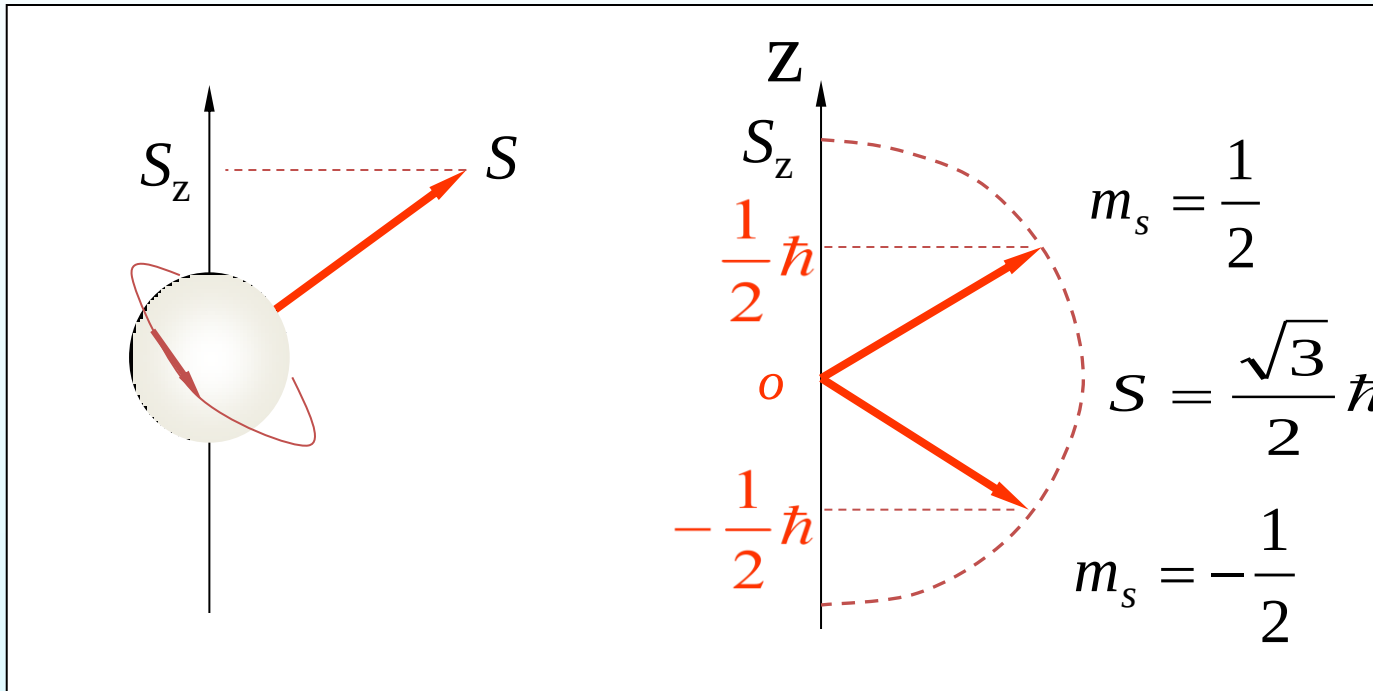


spin magnetic quantum number

$$m_s = \pm \frac{1}{2}$$

Spin angular momentum

$$S_z = \pm \hbar / 2$$



4. Quantum states

We use **four** quantum numbers n, l, m_l, m_s , to denote an state of hydrogen atom.

(1) principal quantum number 主量子数 n , $n = 1, 2, 3, \dots$

大体上决定原子中的电子的能量

(2) orbital quantum number (轨道)角量子数 l , $l = 0, 1, 2, \dots, (n - 1)$

决定电子的轨道角动量, 对能量也有影响

(3) magnetic quantum number 磁量子数 m_l , $m_l = 0, \pm 1, \pm 2, \dots, \pm l$

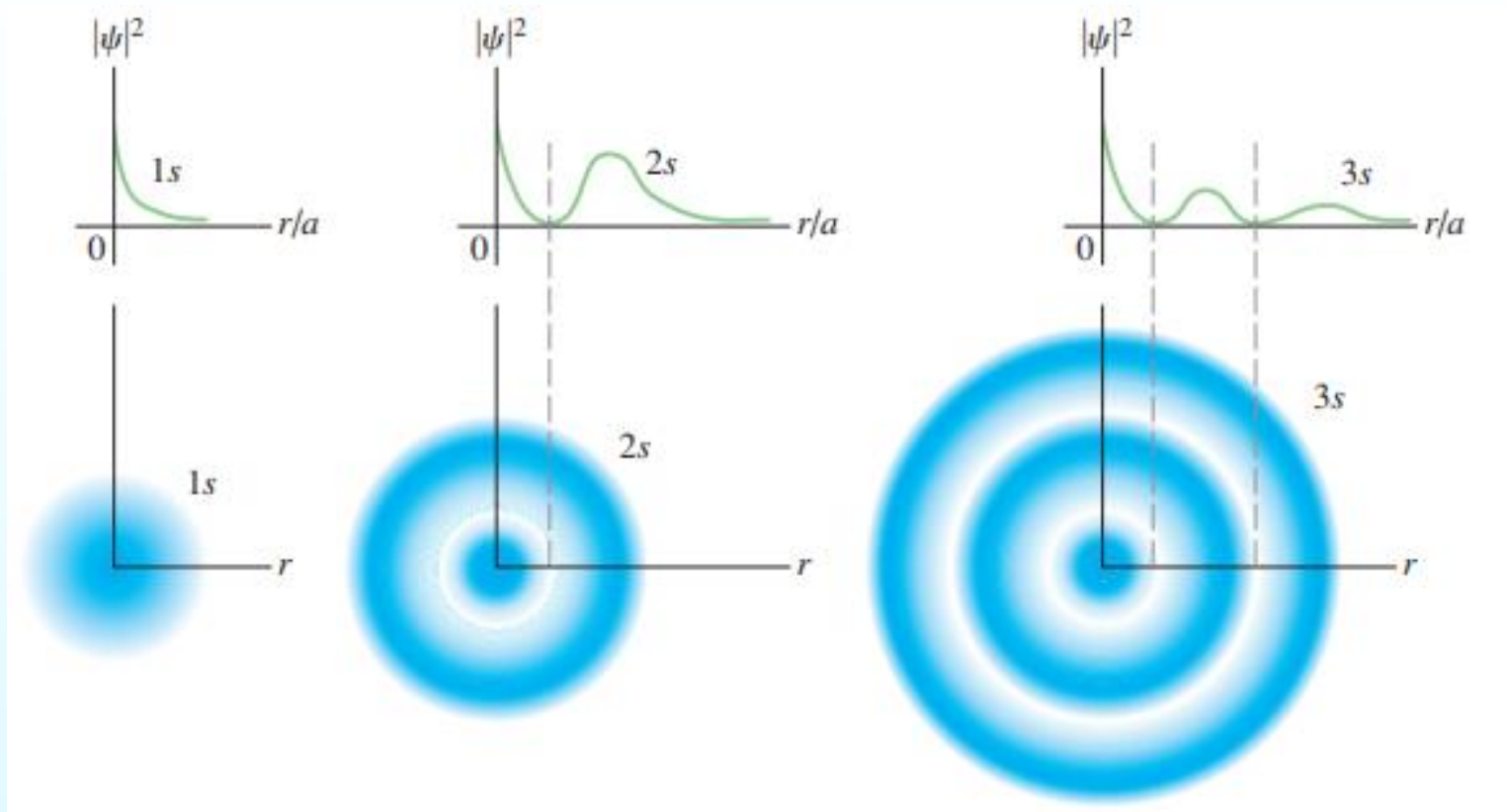
决定轨道角动量在外磁场方向上的分量

(4) spin magnetic number 自旋磁量子数 m_s , $m_s = \pm 1/2$

决定电子自旋角动量在外磁场方向上的分量

Quantum state: (n, l, m_l, m_s)

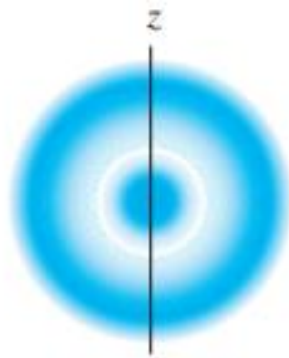
Three-dimensional probability distribution functions for the spherically symmetric 1s, 2s, and 3s hydrogen-atom wave functions.



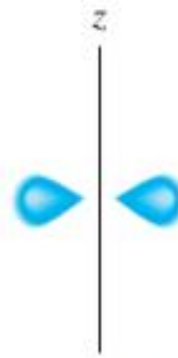
Cross sections of three-dimensional probability distributions for a few quantum states of the hydrogen atom.



$1s, m_l = 0$



$2s, m_l = 0$



$2p, m_l = \pm 1$



$2p, m_l = 0$



$3p, m_l = 0$



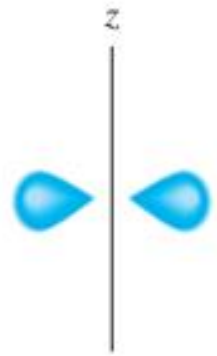
$3p, m_l = \pm 1$



$3d, m_l = 0$



$3d, m_l = \pm 1$



$3d, m_l = \pm 2$

23.6 Periodic table of elements

Electrons with the same value of n are referred to as being in the same shell (主壳层) .

$$n = 1, 2, 3, 4, 5, 6, \dots$$

Demotion: $K, L, M, N, O, P, \dots$ shell

Electrons with the same value of n and l are referred to as being in the same subshell (支壳层)

$$l = 0, 1, 2, 3, 4, 5 \dots$$

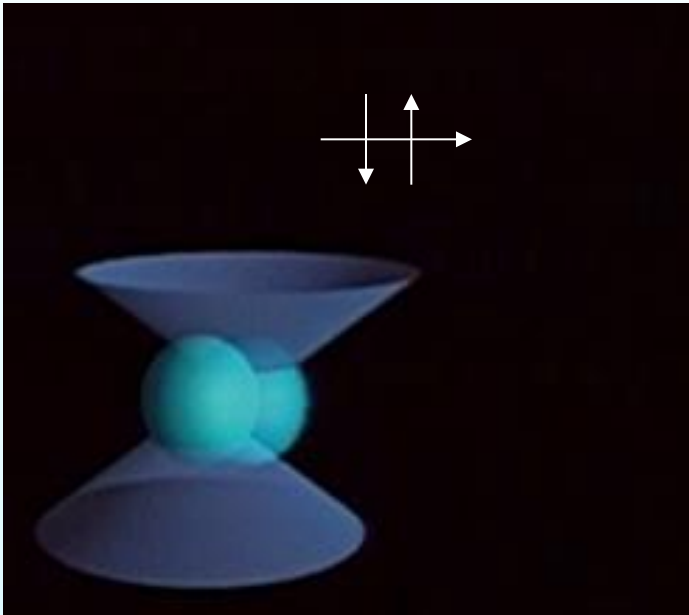
Demotion: $s, p, d, f, g, h \dots$ subshell

原子是由多个电子与原子核组成系统，系统的状态用电子状态分布来描写。用 n 、 l 标记一个电子再指明该态中的电子数——电子组态，若有 x 个电子处于 $n l$ 态，记 $n l^x$

The possible electron states are occupied by the following rules :

(1) **Pauli exclusion principle** 泡利不相容原理

No two electrons in an atom can occupy the **same** quantum state. Thus no two electrons have exactly the same set of quantum numbers n, l, m_l, m_s .



**Wolfgang
Pauli**
(1900-1958)
1945 “Noble”

Building the periodic Table

For each given m_l

The electron also have **two** different spin orientations

For each given l (subshell) 支壳层

the total number of possible electron states $N_l = 2(2l + 1)$

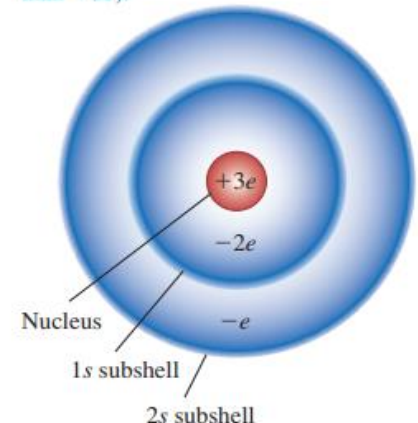
for a given n (shell) 主壳层

the total number of possible electron states

$$N_n = \sum_{l=0}^{n-1} N_l = \sum_{l=0}^{n-1} 2(2l + 1) = 2n^2$$

能级 n 的量子态数为: $N = 2n^2$

On average, the 2s electron is considerably farther from the nucleus than the 1s electrons. Therefore, it experiences a net nuclear charge of approximately $+3e - 2e = +e$ (rather than $+3e$).



原子主壳层和支壳层上最多可能容纳的电子数

<i>l</i> <i>n</i>	0 <i>s</i>	1 <i>p</i>	2 <i>d</i>	3 <i>f</i>	4 <i>g</i>	5 <i>h</i>	6 <i>i</i>	$2n^2$
1 K	2(1s)							2
2 L	2(2s)	6(2p)						8
3 M	2(3s)	6(3p)	10(3d)					18
4 N	2(4s)	6(4p)	10(4d)	14(4f)				32
5 O	2(5s)	6(5p)	10(5d)	14(5f)	18(5g)			50
6 P	2(6s)	6(6p)	10(6d)	14(6f)	18(6g)	22(6h)		72
7 Q	2(7s)	6(7p)	10(7d)	14(7f)	18(7g)	22(7h)	26(6i)	98

(2) Law of minimum energy

An atom is stable if its total energy is a minimum, consequently electrons occupy states with the lowest possible energy.

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p

我国理论科学家徐光宪提出一个经验公式：

$$(n + 0.7 l)$$

该值越大，能级越高。

Electronic configuration of the ground state

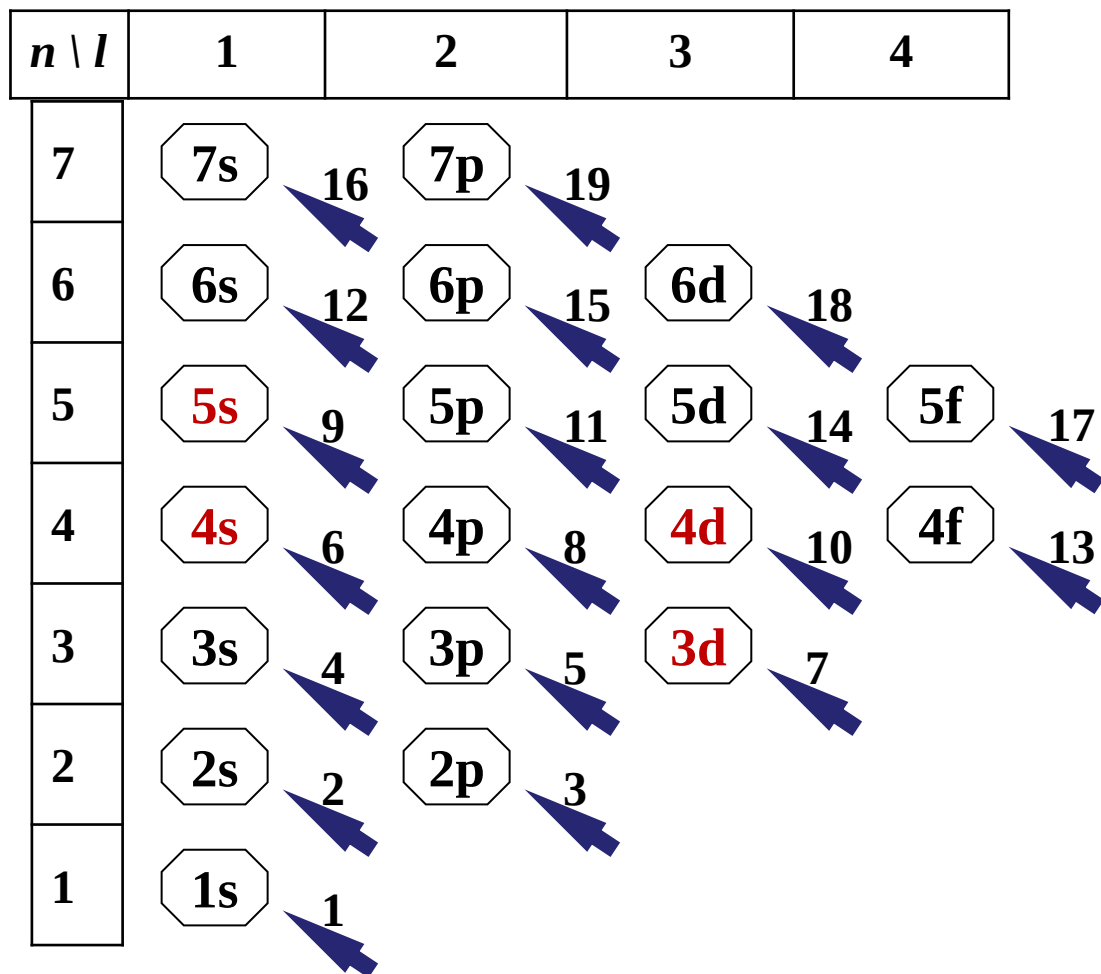
$1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p$

$Z=10: 1s^2 2s^2 2p^6$

$Z=18: 1s^2 2s^2 2p^6 3s^2 3p^6$

$Z=19: 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

$Z=20: 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$



钾K $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^1,$

$4s < 3d$

钾K $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1,$

$1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p \text{ ---}$

H 1 1.008 Hydrogen																	He 2 4.00 Helium
2 IIA												13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	
Li 3 6.94 Lithium	Be 4 9.01 Beryllium											B 5 10.81 Boron	C 6 12.01 Carbon	N 7 14.01 Nitrogen	O 8 16.00 Oxygen	F 9 19.00 Fluorine	Ne 10 20.18 Neon
Na 11 22.99 Sodium	Mg 12 24.31 Magnesium											Al 13 26.98 Aluminum	Si 14 28.09 Silicon	P 15 30.97 Phosphorus	S 16 32.07 Sulfur	Cl 17 35.45 Chlorine	Ar 18 39.95 Argon
		3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8 VIII B	9 VIII B	10 VIII B	11 IB	12 IIB						
K 19 39.10 Potassium	Ca 20 40.08 Calcium	Sc 21 44.96 Scandium	Ti 22 47.88 Titanium	V 23 50.94 Vanadium	Cr 24 52.00 Chromium	Mn 25 54.94 Manganese	Fe 26 55.85 Iron	Co 27 58.93 Cobalt	Ni 28 58.69 Nickel	Cu 29 63.55 Copper	Zn 30 65.39 Zinc	Ga 31 69.72 Gallium	Ge 32 72.61 Germanium	As 33 74.92 Arsenic	Se 34 78.96 Selenium	Br 35 79.90 Bromine	Kr 36 83.80 Krypton

铷Rb, $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^{10}, 4s^2, 4p^6, 4d^1,$

$4s < 3d, 5s < 4f$

铷Rb $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^1$



An electron is in a quantum state for which the magnitude of the orbital **angular momentum** is $6\sqrt{2}\hbar$. How many allowed values of the **z component** of the angular momentum are there?

- A. 7
- B. 8
- C. 16
- D. 17
- E. 20

Example Most **probable electron radius** in hydrogen.
Determine the most probable distance ***r*** from the nucleus at which to find the electron in the ground state of hydrogen.

Solution:

$$\psi_{100} = \frac{1}{\sqrt{\pi r_0^3}} e^{-\frac{r}{r_0}}$$

Wave function for the ground state

$$|\psi_{100}|^2 = \frac{1}{\pi r_0^3} e^{-\frac{2r}{r_0}}$$

Probability density

$$P_r = 4\pi r^2 |\psi|^2$$

Radial probability distribution

For the ground state of hydrogen

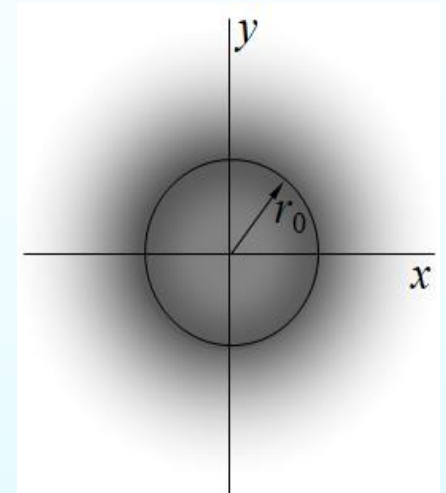
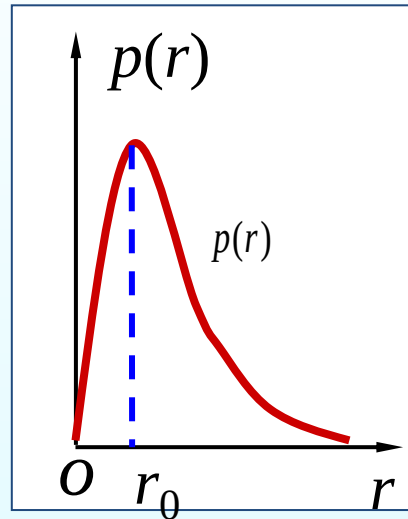
$$P_r = 4\pi r^2 |\psi_{100}|^2 = 4\pi r^2 \frac{1}{\pi r_0^3} e^{-\frac{2r}{r_0}} = 4 \frac{r^2}{r_0^3} e^{-\frac{2r}{r_0}}$$

Determining the most probable electron radius r , take the derivative of P_r with respect to r and set it equal to zero.

$$\frac{dP_r}{dr} = \frac{d}{dr} \left(4 \frac{r^2}{r_0^3} e^{-\frac{2r}{r_0}} \right) = 0$$

$$\left(8 \frac{r}{r_0^3} - \frac{8r^2}{r_0^4} \right) e^{-\frac{2r}{r_0}} = 0$$

$$r = r_0$$



The **most probable distance** r of the electron from the nucleus according to quantum mechanics is at the **Bohr radius**

Example A hydrogen atom is in the **6g state**. Determine (a) the **principal quantum number**, (b) the **energy** of the state, (c) the **orbital angular momentum** (d) the possible values for the **magnetic quantum number**.

Solution: (a) the principal quantum number $n=6$

$$(b) E_n = \frac{E_1}{n^2} = -13.6 \frac{1}{n^2} = -0.378 \quad (eV)$$

$$(c) \quad l = 4, \quad L = \sqrt{20}\hbar$$

$$(d) \quad m_l = 0, \pm 1, \pm 2, \pm 3, \pm 4$$

Label these statements as true or false:

(a) One (and only one) of these subshells can not exist:

$2p, 4f, 3d, 1p$

(b) There are n subshells for each value of n

(c) The smallest value of n for a given value of l is $l+1$

(d) All states with $l=0$ also have $m_l=0$

$\psi(x)$ is the wave function for a particle moving along the x axis. The **probability** that the particle is in the interval from $x=a$ to $x=b$ is given by

(A) $|\psi(b)|^2 / |\psi(a)|^2$

(B) $\int_a^b |\psi(x)|^2 dx$ 🤔

(C) $\int_a^b \psi(x) dx$

(D) $\psi(b) / \psi(a)$

It knows the particle is moving in the one-dimension infinitely deep potential rectangular well, and its wave function is

$$\psi(x) = \sqrt{\frac{2}{a}} \sin \frac{3\pi}{a} x \quad (0 \leq x \leq a)$$

Then, the probability density (概率密度) that the particle appears at the $x = a/6$ is

(A) $\sqrt{2} / \sqrt{a}$

(B) $1/a$

(C) $2/a$

(D) $1/\sqrt{a}$

